Name: _____ **(35 Points)**

Part1 -Multiple Choice Questions

1. The arc length of curve of $r = 5\sin\theta$ from $\theta = 0$ to $\theta = 2$ equals
   a. 5 b. 10 c. 15 d. 20

2. The equation of the leminscate' $r^2 = 4\cos 2\theta$ is equivalent to the rectangular equation :
   a. $\left(x^2+y^2\right)^2 = 4\left(x^2-y^2\right)$
   b. $\left(x^2+y^2\right)^2 = 4$
   c. $\left(x^2+y^2\right) = 4\left(x^2-y^2\right)$
   d. $\left(x^2+y^2\right)^2 = 4\left(x^2-y^2\right)^2$

3. The curve of $r = 8\sin 3\theta$ is a
   a. 6 - petals rose b. 3 - petals rose c. 16 - petals rose d. 8 - petals rose

4. $\int_0^{\frac{1}{2}} \frac{dx}{\sqrt{(1-x^2)^3}} =$
   a. $\frac{1}{\sqrt{3}}$ b. $\frac{1}{\sqrt{5}}$ c. $\frac{1}{\sqrt{15}}$ d. $\frac{1}{\sqrt{2}}$

5. $\int \frac{6x+4}{x^3+4x}\,dx =$
   a. $2\ln|x| + \ln(x^2+4) + 6\tan^{-1}\left(\frac{x}{2}\right) + c$
   b. $\ln|x| + \frac{1}{2}\ln(x^2+4) + 3\tan^{-1}\left(\frac{x}{2}\right) + c$
   c. $\ln|x| - \frac{1}{2}\ln(x^2+4) + 3\tan^{-1}\left(\frac{x}{2}\right) + c$
   d. $2\ln|x| - \ln(x^2+4) + 6\tan^{-1}\left(\frac{x}{2}\right) + c$

6. If the sum of the series $\sum_{k=1}^{\infty} a_k$ is 10, then $\lim_{n\to\infty}\left(a_n + \frac{6n+1}{2n+3}\right) =$
   a. 13 b. 3 c. 4 d. 14

7. The sum of the series $\sum_{k=2}^{\infty} \frac{\sqrt{k+1}-\sqrt{k}}{\sqrt{k^2+k}}$ is
   a. $\frac{1}{\sqrt{3}}$ b. $\frac{1}{\sqrt{2}}$ c. 1 d. not exists(divergent)

8. Which of the following series is absolutely convergent ?

a. $\sum_{k=1}^{\infty}(-1)^k \frac{1}{\ln(k+1)}$

b. $\sum_{k=1}^{\infty}(-1)^k \frac{1}{k^2+1}$

c. $\sum_{k=1}^{\infty}(-1)^k \frac{k}{2k+1}$

d. $\sum_{k=1}^{\infty}(-1)^k \left(\frac{2k+1}{k+1}\right)^k$

9. The radius of convergence of the series $\sum_{k=0}^{\infty} k!(x-1)^k$ is equal to

a. 0 b. 1 c. $\frac{1}{2}$ d. ∞

10. The Maclaurin expansion corresponding to the function, $f(x)=x^3 e^{2x}$

a. $\sum_{k=0}^{\infty}\frac{2^k x^{k+2}}{k!}$ b. $\sum_{k=1}^{\infty}\frac{2^k x^{k+2}}{k!}$ c. $\sum_{k=0}^{\infty}\frac{2^k x^{k+3}}{k!}$ d. $\sum_{k=1}^{\infty}\frac{2^k x^{k+3}}{k!}$

11. The Maclaurin series corresponding to the function $f(x)=\ln(3+x)$ is

a. $\sum_{k=0}^{\infty}(-1)^k \frac{x^k}{(k+1)3^k}$

b. $\sum_{k=0}^{\infty}(-1)^k \frac{x^{k+1}}{k3^{k+1}}$

c. $\sum_{k=0}^{\infty}(-1)^k \frac{x^{k+1}}{(k+1)3^k}$

d. $\sum_{k=0}^{\infty}(-1)^k \frac{x^{k+1}}{(k+1)3^{k+1}}$

12. Find the Taylor series about $x=1$ corresponding to the function, $f(x)=\frac{2x-2}{3-x^2+2x}$

a. $\sum_{k=0}^{\infty}\frac{(x-1)^{2k+1}}{2^{2k}}$ b. $\sum_{k=0}^{\infty}\frac{(x-1)^{2k}}{2^{2k+1}}$ c. $\sum_{k=0}^{\infty}\frac{(x-1)^{2k+1}}{2^{2k+1}}$ d. $\sum_{k=0}^{\infty}\frac{(x-1)^{2k}}{2^{2k}}$

13. (Bessel Function) The domain of Bessel function $J_1(x)=\sum_{k=0}^{\infty}(-1)^k \frac{x^{2k+1}}{2^{2k+1}k!(k+1)!}$ is

a. $(-1,1)$ b. $(-1,0)$ c. $(0,1)$ d. $(-\infty,\infty)$

14. If $y=A\cosh 2x+B\sinh 2x$ satisfies the equation, $y''+y=10e^{-2x}$ then the orderd pair $(A,B)=$

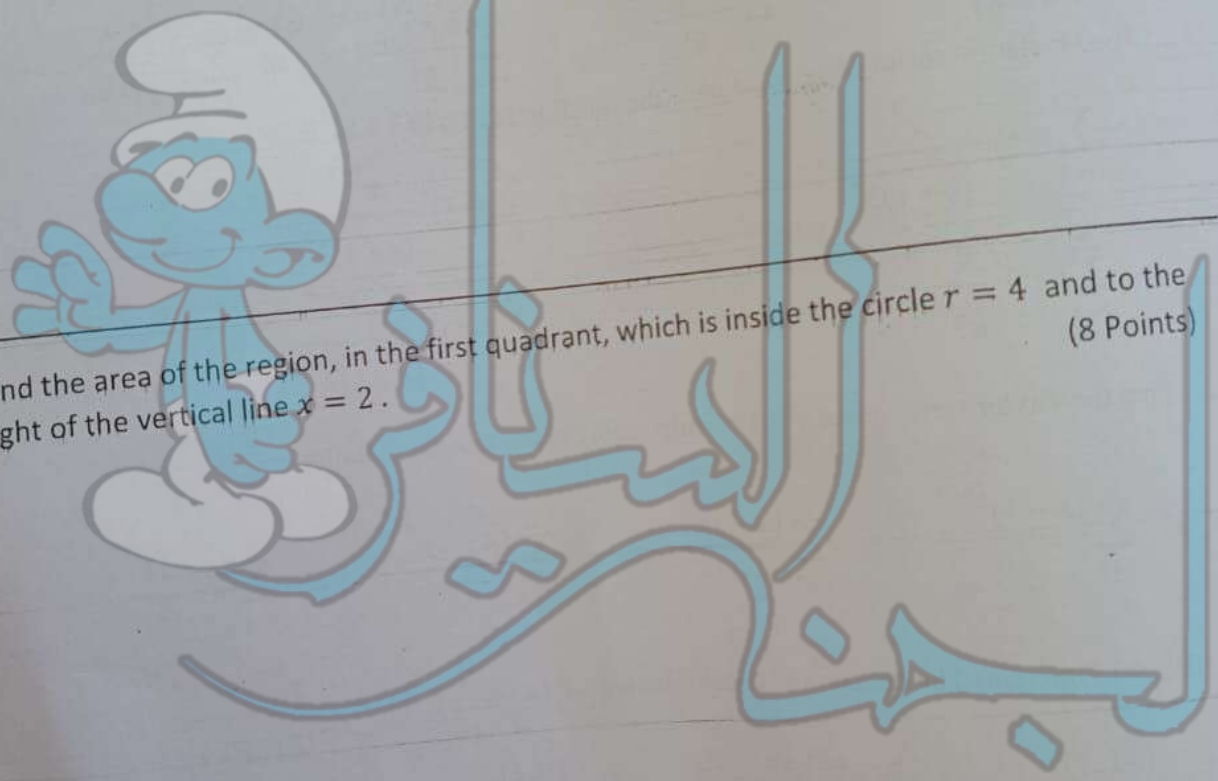
a. $(2,2)$ b. $(-2,-2)$ c. $(-2,2)$ d. $(2,-2)$

(15 Points)

Part 2 -Essay Questions

(7 Points)

1. If $f(x) = \begin{cases} 0 & , \ 0 \le t < 2 \\ t & , \ t \ge 2 \end{cases}$, find $\int_0^\infty f(t)\, e^{-st}\, dt \quad s > 0$



2. Find the area of the region, in the first quadrant, which is inside the circle $r = 4$ and to the right of the vertical line $x = 2$. (8 Points)

GOOD LUCK